

Catch every strip defect where it is born, not where it is shipped

AI surface inspection that finds, grades and routes every stainless strip defect to the process that caused it.

90%

of defects caught on real mill strip it had never seen

4.7%

of clean strip wrongly flagged, fewer than 1 in 20

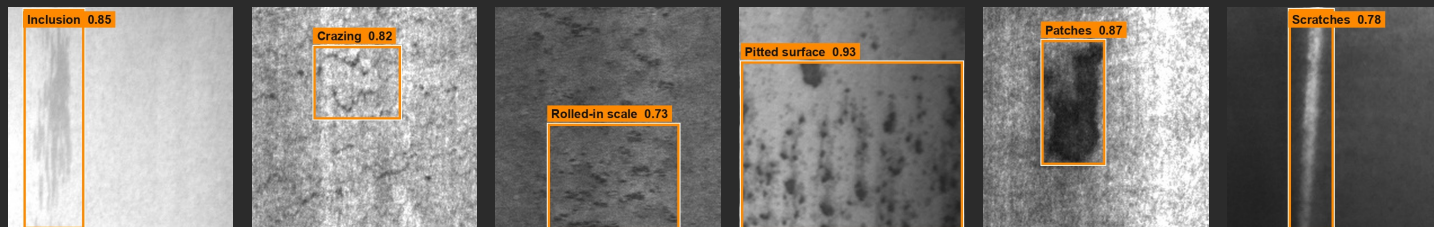
~₹6.7 cr

a year at stake per line, at a 1% downgrade rate

7 months

to shadow mode on one line, go / no-go at month 7

REAL OUTPUT FROM OUR SYSTEM, ON STEEL IMAGES IT HAD NEVER SEEN



TEAM Futuristic | Krishna Kanta Mondal · Prathmesh Walimbe | IIT Bombay



TRY THE LIVE DEMO

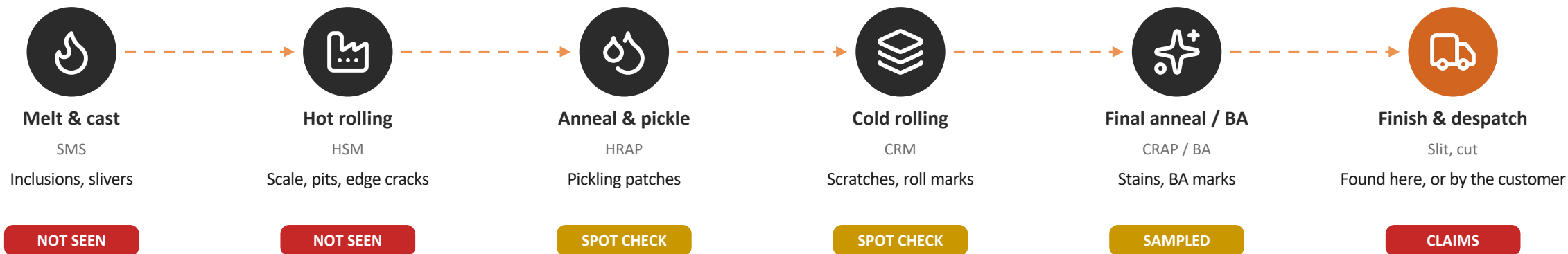
surface-vision.github.io

Runs in your browser. Nothing leaves the page.

A defect born at the caster is paid for at every later stage, but today it is found only at the end

Where stainless surface defects are born, and where they are seen today

VALUE ADDED KEEPS RISING →

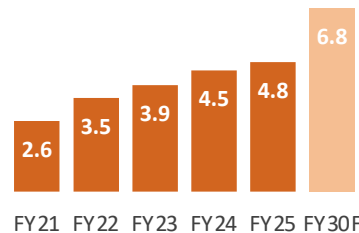


Why it hurts more on stainless

- 300 m/min**
too fast for the eye to see the whole surface
- ₹126k-165k**
of nickel in every tonne of 304, given away on a downgrade
- No repair**
damage on 2B/BA mill finishes cannot be polished out

Why now

India demand, Mt +84% in 5 yrs



- 1.73 Mt imports**
FY25; anti-dumping probe on cold-rolled 300/400, Sep 2025
- IS 6911 mandatory**
BIS certification for sheet and strip, since Aug 2024
- 2.67 MTPA**
JSL cold rolling by FY28, up from 2.05

Cost of quality: where AI moves the money

spend moves from failure to prevention

Prevention defect sent to its owner in minutes	Internal failure caught before more value is added
Appraisal 100% of the surface, every coil	External failure fewer escapes to customers

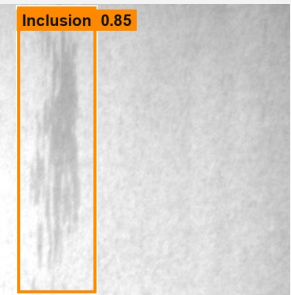
Poor quality costs ~15% of sales in manufacturing (IISE).

Sources: AMETEK/Ternium (2020); ASSDA; IISE; ASTM A240; LME nickel USD 16,402/t at INR 95.8/USD (22 Sep 2026); ISSDA (FY25); DGTR (29 Sep 2025); BIS Steel QCO 2024; JSL.

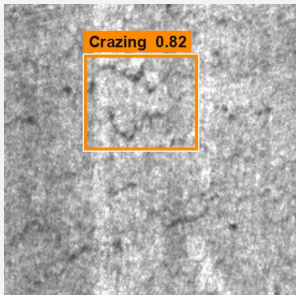
The problem is not that defects exist. They are found late, by sampling, and nobody is told which process caused them.

On stainless the surface is the product, so every defect type must reach the process that caused it

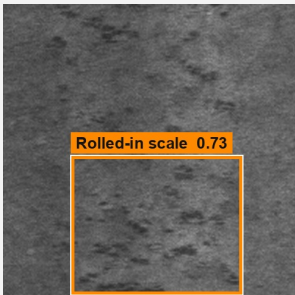
Six defect families, each traced to the process that caused it (real output from our system)



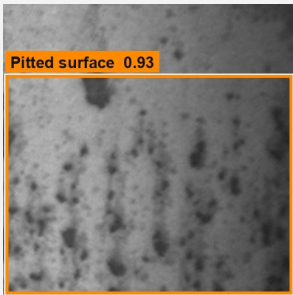
Born: **Caster**
Owner: **SMS**
Fix: Tundish level, mould flux



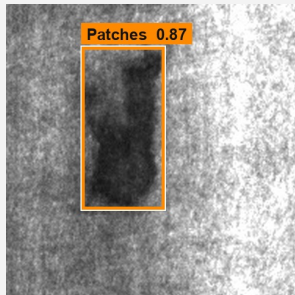
Born: **Reheat, hot rolling**
Owner: **HSM**
Fix: Reheat and coiling temperature



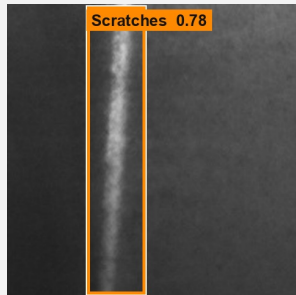
Born: **Hot rolling**
Owner: **HSM**
Fix: Descaler pressure, nozzles



Born: **Reheat, pickling**
Owner: **HSM / pickling**
Fix: Furnace atmosphere, over-pickling



Born: **Anneal & pickle**
Owner: **Pickling line**
Fix: Acid strength, line speed



Born: **Cold rolling, handling**
Owner: **CRM / logistics**
Fix: Guides, rolls, coil handling

Stainless-only types to add with JSL data:

Roping

Ridging (ferritic 430)

Orange peel

Anneal colour

BA stains

Lamination

Precedent: who already does this, and what we copy

Outokumpu	Europe's #1 stainless maker; AI inspection live at Tornio since 2024, runs offline. Copy: offline edge, multi-site
Tata Steel	Kalinganagar (WEF Lighthouse) finds cold-rolled surface defects by video analytics. Copy: in-house team owns it
Parsytec	On stainless lines: 170 µm cameras, bright + dark-field light, a decision per coil. Copy: optics, coil decision

Grade-by-defect risk: which defects to train for first

Grade	Slivers	Scale	Scratches	Edge cracks	Ridging
200 series (Cr-Mn)	M	M	H	M	L
300 series (304, 316)	H	M	H	L	L
400 series (409, 430)	H	M	M	M	H
Duplex (2205)	M	H	M	H	L

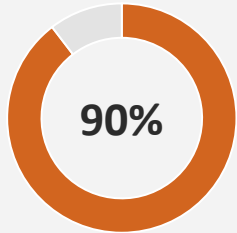
OUTPUT
300 series is the pilot grade: highest exposed-finish risk and the most nickel per tonne.

Images: held-out test frames, boxes drawn by our system. Sources: Nortal/Outokumpu (2024); Tata Steel; Parsytec (Stainless Steel World); Leão 2021; Steel in Translation 2023. Owners: team hypothesis. Grade risks: team hypothesis.

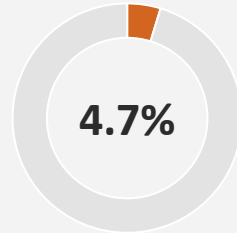
Classify, don't just alarm: a typed defect becomes a work order for a named owner. The pilot belongs on 300-series cold-rolled strip.

Four design choices that make the system work on a real line

1 Alarms you can act on



of defects caught
258 of 288 defective mill
images



of clean strip flagged
14 of 300 clean mill
images

So we: prove it on JSL's own clean and defective strip before any alarm is trusted.

2 One verdict per coil, not a stream of alarms

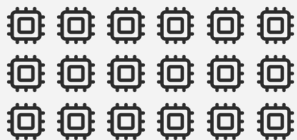


All defects on a coil roll up into one decision, using limits set per grade and finish. **Example: a light scratch can pass on 2B for tubes but downgrade BA for appliance panels.**

So we: agree accept, downgrade and hold limits per grade and finish with JSL Quality in P0.

3 Image detail sets what is seen, and the hardware bill

0.2 mm per pixel



~18 processors

sees defects from ~0.5 mm

0.5 mm per pixel



~3 processors

sees defects from ~1.3 mm

1.28 mm per pixel



1 processor

fine scratches vanish (~3 mm)

1.28 m strip at 250 m/min. Processors scale with pixels; a defect needs 2-3 pixels.

So we: fix the image resolution with JSL first, then size the hardware.

4 Operators act only on alarms they can trust



Every alarm shows its evidence

photo, defect type, severity and position on the coil



Four weeks of shadow mode first

it only advises; automatic holds start at ≥ 90% operator agreement



Every override is reviewed

weekly with the line and quality; the system improves from it

So we: earn trust on the shift floor before the system is allowed to act.

Results measured by us on public steel data the system had never seen. Hardware sizing derived from our line-speed benchmark.

Success is decided on the line, not in the lab: JSL's own strip, the right image detail, coil-level rules and operator trust.

Detect on the strip, decide at the coil, route to the owner, and learn from every correction

How it works: six stages from camera to corrective action



Capture

Both faces, bright + dark-field light, 0.2 mm detail

TO BUILD

At the line: cameras, lights



Check

Is it usable steel? If not, skip it

BUILT



Detect

Find and classify every defect

BUILT

Edge server beside the line, runs offline



Score

Confidence, severity 0-100, coil position

BUILT



Decide

Accept, downgrade or hold, per grade

PARTIAL

Plant network (IEC 62443)



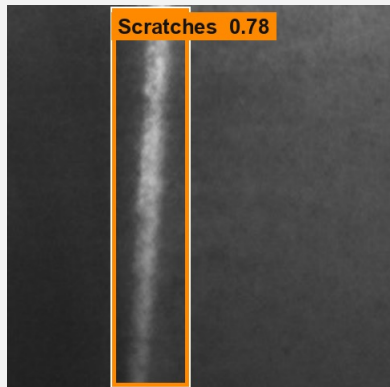
Act & learn

Alert, hold, work order; operator feedback improves it

TO BUILD

People: screen, work orders

Running today



Upload a strip image, get:

- box on the defect
- defect type
- confidence
- severity score

16 samples built in, including clean mill strip. Or drop your own photo.

surface-vision.github.io runs in the browser

What JSL gets that a standard system does not



Alarms you can act on

only 4.7% of clean strip raises one



Root-cause routing

every alarm becomes a work order for its owner



One verdict per coil

accept, downgrade or hold, by grade and finish



JSL owns the system

its data and software; no vendor lock-in

Build vs buy

	Cost per line	Proven at speed	Tuned to JSL grades	JSL owns the system
Commercial proven; the vendor owns it	●	●	●	●
Fully in-house JSL owns it; slow to prove	●	●	●	●
Hybrid (recommended) vendor cameras, JSL software	●	●	●	●

● strong
● partial
● weak

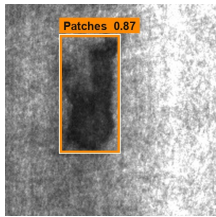
Output: hybrid is never weak.

BUILT = working in our prototype today, live at surface-vision.github.io.

Buy proven cameras, own the software: vendor-grade capture, tuned to JSL's grades and wired to JSL's process owners.

It already works on public steel data; the pilot's job is to prove it on JSL's own strip

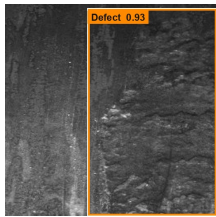
Tested on three public steel datasets



NEU-DET

1,800 images, 6 types

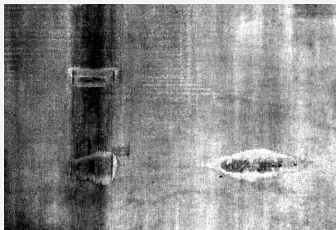
University benchmark, hot-rolled strip



Severstal

12,568 real mill frames

Production strip from a steel mill

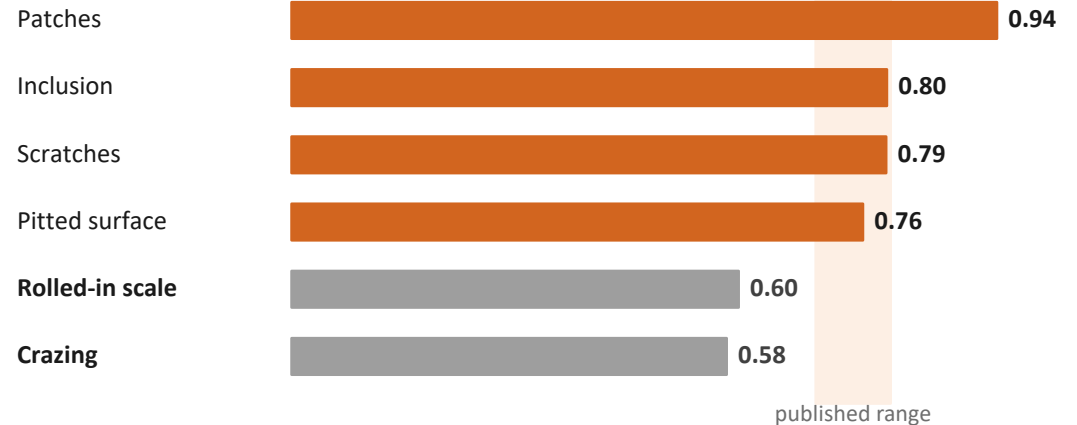


GC10-DET

2,294 frames, 10 types

Full-width line-scan camera frames (roll marks shown)

How well it finds each defect type (accuracy, held-out images)



Headline results



178 of 180

test images: the right defect type named

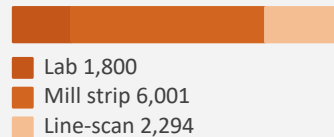


Live

anyone can test it now at surface-vision.github.io



~10,100



Known weak spots, and the fix for each

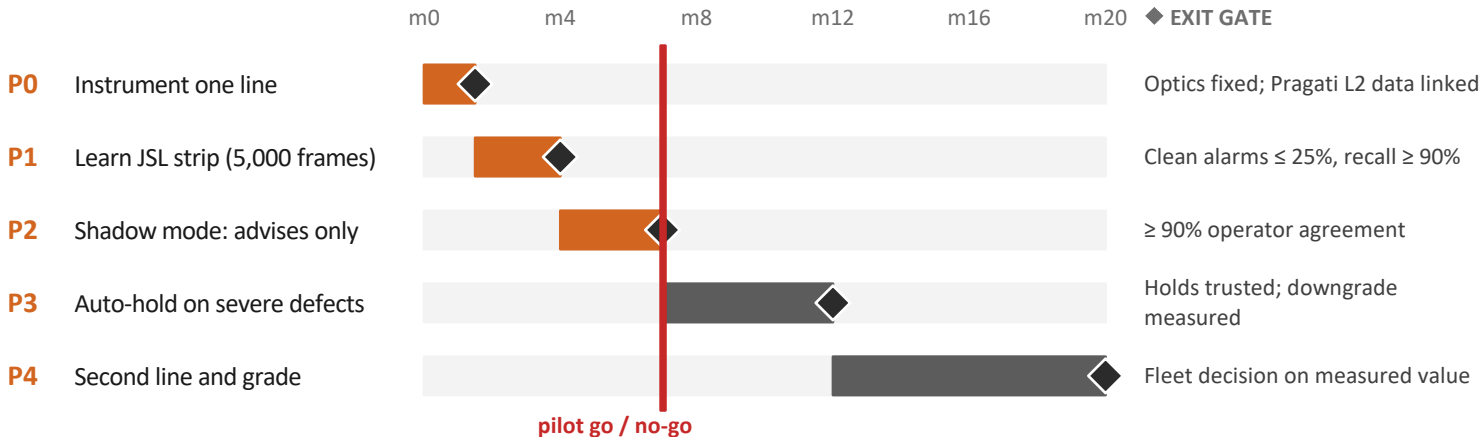
Weak spot	Today (0-1)	Fix in the pilot
Low-contrast texture: crazing, rolled-in scale	0.58-0.60	Angled lighting; JSL labels in P1
Real mill frames are harder than the lab benchmark	0.57 vs 0.75	Prove it on JSL's own strip in P1
Roll marks and edge cracks: few examples	0.20	Edge cameras; targeted labelling with JSL

Datasets: NEU-DET (Northeastern Univ.); Severstal steel defects (Kaggle); GC10-DET (Lv et al. 2020, CC BY 4.0). All scores on held-out images the system had never seen.

Strong on four of six defect families today. Every weak spot is known, and each has a fix in the pilot plan.

Seven months to shadow mode on one line, and nothing touches the mill until trust is earned

Stage-gate roadmap (months)



Pilot scope: in and out

IN THE PILOT

- One 300-series cold-rolled line, both faces
- 2B and BA finishes
- Six defect families, plus unclassified
- Read-only until P3; one write path

LATER

- Hot end and caster-side inspection
- Edge cracks: needs edge cameras
- Automatic process correction
- Other plants: after P4

Decision owner: **JSL Quality** Integration: **Pragati team**

Top risks, and when each is retired

Risk	Mitigation	Retired
Misses on JSL's own strip at first	Learn from JSL clean and defect images	P1
Mirror-like BA/2B finish hides defects	Bright + dark-field light; deflectometry for BA	P0-P1
Rare defects missed	Edge cameras; targeted labelling	P1-P3
Operators ignore alarms	Evidence on every alarm; shadow mode first	P2
Plant network, heat and fumes	IEC 62443 zones, runs offline; sealed enclosures	P0
Quality slips after an update	Each update must beat the live system first	P2+

Pilot KPIs: how JSL will know it works

$\geq 90\%$

defective strip flagged
measured 258 of 288 (89.6%)

$\leq 25\%$

clean strip wrongly flagged
measured 4.7% on mill strip

100%

coils with a defect map
design target

< 0.2 s

defect to operator screen
design target

$\geq 90\%$

operator agreement, shadow
measured in P2

↓ vs today

downgrade rate on the line
measured in P3

Pragati: JSL's digitalisation programme; Phase 2 brings Level-2 process data at Jajpur. KPI targets to be agreed with JSL Quality in P0.

Commit one line for seven months. The month-7 go/no-go is decided on JSL's own alarm and recall numbers.

At a conservative 1% downgrade rate, each line has ~₹6.7 cr a year at stake; ~₹43 cr across the fleet

Loss pool per line, ₹ cr a year

$0.4 \text{ Mt} \times \text{downgrade rate} \times \text{price discount} \times ₹167,407/\text{t}$

Downgraded	10% off	15% off	20% off
0.5% of volume	3.3	5.0	6.7
1.0% of volume	6.7	10.0	13.4
2.0% of volume	13.4	20.1	26.8

Base case (dark): ₹6.7 cr a year per line; ~₹43 cr across JSL's 2.57 Mt. Non-prime coil sells 10-30% below prime, so these discounts are conservative.

Value levers, and how the pilot measures each

Value lever	How the system creates it	Measured in the pilot as
Fewer downgrades	defects caught upstream, before more value is added	downgrade rate by line and grade
Fewer customer claims	a defect map travels with every coil	claims per 1,000 t shipped
Less rework	no re-pickling or re-grinding of defects found late	rework tonnes per month
Faster root-cause fixes	each alarm reaches the process that caused it	hours from defect to corrective action
Premium orders	documented surface quality for rail, appliances, exports	share of exposed-finish orders

Impact beyond the P&L



Coverage

100% of both faces, every coil

today: spot checks



Customers

Defect map per coil for rail, metro, auto buyers

Vande Bharat supplier



Process

Defects tied to heat and coil IDs

6 families, each owned



Safety

Severe-defect holds prevent strip breaks

Ternium's business case



Sustainability

Less rework and remelt; backs net zero

JSL: 1.76 tCO2e/t, FY26



People

Inspectors move to decisions

HR AI: 95% self-served

WHAT WE ASK OF JSL

- One 300-series cold-rolled line, 7 months
- Vendor quotes for cameras and lighting in P0
- Downgrade rate by line and grade
- A metallurgist at 20% for root causes

In return: the measured downgrade rate, the value captured, and a go / no-go at month 7.

Inputs: JSL FY26 (INR 167,407/t; 2,565,902 t sold). Base case: a 0.4 Mt line, 1% downgraded at a 10% discount. Value at stake, not a savings forecast: JSL's own data sets the real figure.

Recommendation: approve a 7-month pilot on one 300-series cold-rolled line. JSL's own downgrade rate decides fleet rollout.